MPG No. 13.1.44 $4mm$ [Type I, tetragonal] [T tensor]

* Rank 0 tensor.

$$\begin{bmatrix} T \end{bmatrix}$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & T_z \end{bmatrix}$$

$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & T_{xx} & 0 \\ 0 & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)}$$
$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{xxz} \\ 0 & 0 & T_{zzz} \\ 0 & T_{xxz} & 0 \\ T_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)}$$
$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{T}_z^{(2)}$$
$$\frac{T_{xxz}}{2} = \mathbb{T}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ -T_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = \mathbb{T}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxzz} & 0 & 0 & 0 \\ T_{xxzz} & T_{xxzz} & T_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)}$$
$$T_{xxzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)}$$
$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)}$$
$$T_{xyxy} = \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & T_{xxzz} & 0 & 0 & 0 \\ 0 & 0 & T_{xxzz} & 0 & 0 & 0 \\ -T_{xxzz} & -T_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxzz} = 3\mathbb{T}_u^{(3)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & 0 \\ 0 & T_{yzyz} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)}$$
$$T_{xyxy} = \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzyz} & 0 & 0 \\ 0 & -T_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzyz} = -3\mathbb{T}_u^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzyz} = -3\mathbb{T}_u^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{zzxx} & T_{zzxx} & T_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzzy} & 0 & 0 & T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} \\
T_{xxyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} \\
T_{xxzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} \\
T_{zzxx} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} \\
T_{zzzz} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{yzyz} &= \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} \\
T_{yzzy} &= \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} \\
T_{xyxy} &= \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{yzzy} & 0 & 0 & -T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & -T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} \\
T_{yzzy} &= -\mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} \\
T_{xyxy} &= \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & 0 & 0 \\ 0 & T_{xxxx} & 0 \\ 0 & 0 & T_{zzzz} \\ 0 & 0 & T_{yyzz} \\ T_{zzxx} & 0 & 0 \\ 0 & T_{xxyy} & 0 \\ 0 & T_{yzzy} & 0 \\ 0 & 0 & T_{zxxz} \\ T_{xxyy} + T_{yyzz} - T_{yzzy} - T_{zxxz} + T_{zzxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} \\
T_{zzzz} &= 3\mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{yyzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} \\
T_{zzxx} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} \\
T_{xxyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} \\
T_{yzzy} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} \\
T_{zxxz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)}
\end{aligned}$$